Flowcharts & Visuals

Engineering Chemistry (UE26CY151A) · Semester 1 · Unit 1 · Compiled from PESU AV-summary transcripts, class slides, the Unit-1 slide PDF and past ESA papers · 13 Oct 2026

★★★ high · ★★ medium · ★ low/conceptual weight | Not on current slides Transcript only Added for clarity | source tags: [S004·8] slide, [T005] transcript, [PDF] 137-page deck, [ESA] past paper

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M1 · Molecular Spectroscopy

T1 · EM radiation & interaction with matter

Atomic vs molecular spectra

	Atomic	Molecular
Transition	between atomic orbitals	HOMO→LUMO + vib + rot
Appearance	line spectra	band spectra

T2 · Electromagnetic spectrum & regions

Microwave vs IR vs UV-Vis

	Microwave	IR	UV-Vis
Transition	rotational	vibrational	electronic
Condition	permanent dipole	dipole changes	any molecule
Rule	$\Delta J = \pm 1$	$\Delta v = \pm 1$ (SHO)	none for v; Franck–Condon
Gives	B, I, r_o	k, $\bar{\nu}$	electronic states

Rayleigh vs Raman

	Rayleigh	Stokes	Anti-Stokes
Frequency	unchanged	lower	higher
Molecule	unchanged	gains energy	loses energy

T3 · Born–Oppenheimer approximation

—

T3b · Beer–Lambert law

—

T4 · Rotational (microwave) spectroscopy

F3 · Spectroscopic activity checker

Permanent dipole? → Yes: microwave (rotational) active

↓

Dipole changes during a vibration? → Yes: IR active

↓

All molecules → electronic (UV-Vis) active

↓

HCl: all three · CO: all three · H_2, Cl_2: electronic only · CO_2: IR (asym. stretch, bend) + electronic, not microwave · H_2O: all three · HI: all three

F4 · Rotational spectrum → bond length

Read first line (or spacing) → B = line/2

↓

$I = h / (8\pi^2 cB)$

↓

$\mu = m_1 m_2/(m_1+m_2)$ in kg

↓

$r_0 = \sqrt{(I/\mu)}$

T5 · Vibrational (IR) – SHO

F5 · IR line → force constant (and reverse)

$\bar{\nu} \rightarrow k = 4\pi^2 c^2 \bar{\nu}^2 \mu$

↓

$k \rightarrow \bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)} \rightarrow ZPE = \frac{1}{2}\bar{\nu} \rightarrow \varepsilon_v = (v+\frac{1}{2})\bar{\nu}$

T6 · Vibrational – anharmonic

F6 · Anharmonic solver

fundamental = $\bar{\omega_e}(1-2x_e)$; overtone = $2\bar{\omega_e}(1-3x_e)$

↓

ratio overtone/(2·fund) = $(1-3x_e)/(1-2x_e) \rightarrow x_e$

↓

$\bar{\omega_e}$ = fundamental/$(1-2x_e)$

SHO vs anharmonic

	SHO	Anharmonic
Potential	parabola	Morse
Levels	equal spacing	converge

	SHO	Anharmonic
Rule	$\Delta v = \pm 1$	$\Delta v = \pm 1, \pm 2, \pm 3 \ldots$
Spectrum	1 line	fundamental + overtones + hot bands

T7 · Electronic spectroscopy & Franck–Condon

F7 · Franck–Condon outcome

Compare r_e' with r_e''

↓

Equal → (0,0) strongest

↓

Longer → (v′,0), v′>0 strongest

↓

Much longer → continuum (dissociation)

M2 · Phase Equilibria

T8 · Basic terms & phase rule

F9 · Phase-rule application

Count P

↓

Count C (minimum constituents)

↓

Gas absent & constant pressure? → F=C−P+1, else F=C−P+2

↓

Classify F=2 bivariant, 1 univariant, 0 invariant

T9 · Derivation of phase rule

F8 · Phase-rule derivation

Variables = P(C−1)+2

↓

Constraints = C(P−1)

↓

F = variables − constraints = C−P+2

T10 · One-component system: water

—

T11 · Two-component: reduced phase rule & Pb–Ag

F10 · Pb–Ag cooling path

Melt cools → Pb separates, melt gets richer in Ag along curve to O

↓

At O (303 °C): Pb + Ag + melt, F=0

↓

Below O: solid Pb + solid Ag

T12 · Thermal analysis / cooling curves

F11 · Thermal analysis → phase diagram

Cool melts of several compositions

↓

Record T vs time

↓

Read break & halt temperatures

↓

Plot vs composition → diagram

M3 · Computational Chemistry

T13 · Computational chemistry

F12 · Computational methods tree

Classical: molecular mechanics, molecular dynamics (fast, lower accuracy)

↓

Quantum: semi-empirical → DFT → ab initio (slower, more accurate)

F13 · Ethylene electronic structure

Define geometry

↓

Select basis set

↓

Apply method

↓

Solve equations → wavefunction/density

↓

Visualise HOMO, LUMO

F14 · AI-enhanced workflow

Quantum simulations

↓

Dataset

↓

Train ML model

↓

Predict properties in ms

MM / MD / semi-empirical / ab initio / DFT

Method	Basis	Speed	Accuracy
MM	classical springs	fastest	lowest
MD	laws of motion	fast	low
Semi-empirical	Schrödinger, approximated	fast	medium
DFT	electron density	medium	high
Ab initio	wavefunction	slowest	highest